Divyanshu Chauhan

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PROFESSIONAL SUMMARY

Full Stack AI Developer with experience in building production-grade RAG systems, LLM-powered applications, and scalable backend architectures. Skilled in designing multilingual AI workflows, hybrid retrieval systems, and async processing pipelines using Python, FastAPI, Redis, Celery, and vector databases. Proven impact in improving retrieval quality, reducing latency, and optimizing AI inference cost through practical engineering decisions. Passionate about Agentic AI, LangChain, LangGraph, and real-world GenAI applications.

CORE COMPETENCIES

Generative AI • RAG Architecture • LLM Applications • Agentic AI • LangChain • LangGraph • FastAPI • Python Backend Development • Vector Databases • pgvector • Hybrid Search (BM25 + Dense Retrieval) • Redis • Celery • AI System Design • Prompt Engineering • Docker • CI/CD • AWS • REST APIs • Event-Driven Architecture • NLP • RFID/NFC Systems

TECHNICAL SKILLS

- **AI/ML & GenAI:** RAG (Naive/Simple/Advanced), LLM Integration, Prompt Engineering, Embeddings, Vector Search, Hybrid Search (BM25 + pgvector), LangChain, LangGraph, Agentic AI, Ollama, Groq API
- **Backend & System Design:** FastAPI, Flask, Node.js, REST API Design, Async Processing, Microservices, Event-Driven Architecture
- **Programming Languages:** Python, SQL, Bash, Java, JavaScript
- **Databases & Storage:** PostgreSQL, pgvector, MySQL, MongoDB, Redis (cache + message broker), Celery (task queue)
- **Cloud & DevOps:** AWS (EC2, S3), Docker, Jenkins, GitLab CI, CI/CD Pipelines, Infrastructure Automation
- **Observability & Tools:** Prometheus, Grafana, ELK Stack, Postman, Git/GitHub, Agile/Scrum, Jira

EXPERIENCE

Full Stack AI Developer (Python + AI), Infosware Private Limited | Noida, India — Mar 2026 – Present

- Architected a production-grade White-Label RAG Chatbot SaaS, designing the complete pipeline from PDF/OCR ingestion → chunking → BGE-M3 embeddings → pgvector storage → hybrid retrieval (BM25 + dense search + RRF re-ranking) → LLM response generation using Ollama (Llama 3.1 8B).
- Improved RAG retrieval accuracy by 30–40% by implementing three parallel pipelines (Naive, Simple, Advanced) with shared query routing while maintaining P95 latency under 2 seconds.
- Debugged hybrid retrieval failures caused by BM25 index staleness after document updates and implemented an event-driven Celery re-indexing system, reducing false retrievals by 60%.
- Designed scalable async processing architecture using Redis + Celery, reducing document processing time by 50% and enabling horizontal scaling.
- Reduced AI inference cost by 25% using prompt compression, Redis response caching, and query deduplication identified through Prometheus metrics.
- Implemented multilingual support for 10+ Indian languages using AI4Bharat Indic LID + IndicTrans2, expanding accessibility for WhatsApp-based chatbot systems.
- Built and integrated an RFID/NFC smart access system supporting secure read/write, password protection, and lock operations for NTAG213/216 tags with real-time backend validation workflows.

QA Software Engineer, Tekshapers (Magic EdTech) | Noida, India — Apr 2024 – Mar 2026

- Designed and built a Python-based NLP pipeline to automate ALT-text generation for EPUB accessibility content — reduced manual validation effort from 3 days to 2 hours (95%-time reduction). Architecture decision: rule-based classifier first, ML model only for edge cases — kept it interpretable for compliance audit trails.
- Debugged flaky QTI test failures in production by tracing GUID mismatches across assessment launch workflows — identified a race condition in concurrent record lookup; patched and documented the fix, preventing recurrence across 3 product teams.
- Built automated regression suite for SPOUT (Standards at Point of Use) tool validating state standards mapping (GA 2023, ME 2020) — improved testing efficiency by 40% and reduced defect leakage by 30%.
- Ensured 99.9% system reliability during release cycles through EDC closure validation and pre-release deployment checks — received formal client appreciation for zero critical post-release defects.

DevOps Intern, AR Electricals | Noida, India — Feb 2024 – Aug 2024

- Designed and implemented Jenkins CI/CD pipelines from scratch for multi-environment deployments — reduced deployment time by 70% (from ~45 min to ~13 min). Tradeoff: chose Jenkins over GitHub Actions to avoid vendor lock-in given on-premise infra requirements.
- Automated build, test, and deployment workflows using Bash + Docker — eliminated 50% of manual deployment steps and standardized environment configuration across dev/staging/prod.

PROJECTS

1. RAG AI Chatbot SaaS Platform | *Python, FastAPI, pgvector, Redis, Celery, Ollama, LangChain* 2026

- Challenge: Indian SMBs needed a multilingual AI chatbot but couldn't afford per-query GPT-4 costs at scale. Solution: architected a self-hosted RAG platform with local LLMs (Ollama/Llama 3.1 8B) eliminating per-token API costs entirely.
- System design decision: implemented three RAG pipelines (Naive for speed, Simple for accuracy, Advanced with re-ranking for quality) behind a unified API — clients get SLA-based routing without knowing the internals.
- Technical challenge faced: chunk boundary problem — semantic search was splitting context mid-sentence. Solved using parent-child chunking strategy: small child chunks for retrieval precision, large parent chunks for LLM context quality.
- Outcome: 35% improvement in response relevance, 50% reduction in processing time, −40% token cost vs GPT-4 baseline. Learned that retrieval quality matters more than LLM size for domain-specific Q&A.

2. RFID/NFC Smart Access System | *Python, REST APIs, RFID/NFC, NTAG213/216* **2026**

- Developed an RFID/NFC-based smart access system enabling secure authentication using read/write, password protection, and lock operations for NTAG213/216 tags.
- Designed backend validation workflows to verify access permissions in real time and trigger automated access decisions.
- Solved NFC write inconsistencies caused by authentication and lock-state conflicts through memory-page validation and error handling.
- Improved operational reliability by implementing secure tag validation and preventing unauthorized access attempts.

3. CI/CD Deployment Pipeline | *Flask, Express, AWS EC2/S3, Docker, Jenkins* 2025

- Problem: deployment took 30 minutes of manual steps with high error rate. Designed an automated pipeline with build → test → dockerize → deploy stages.
- Tradeoff decision: used blue-green deployment strategy over rolling updates to achieve zero-downtime releases — accepted the cost of running dual environments for the reliability guarantee.
- Result: 80% reduction in deployment time (30 min → 5 min), zero failed deployments in production post-implementation. Learned that deployment speed and safety are not mutually exclusive with the right architecture.

EDUCATION

Master of Computer Applications – Gurukul Kangri Vishwavidyalaya Haridwar | CGPA: 8.62

Relevant Coursework: Advanced Algorithms, Software Engineering, Computer Networks, Database Management, AI/ML Fundamentals

Bachelor of Science (Computer Science) – Gurukul Kangri Vishwavidyalaya Haridwar | CGPA: 8.46

Relevant Coursework: Data Structures, Cloud Computing, DBMS, Linux, Software Development

CERTIFICATIONS

- IBM RAG and Agentic AI Professional Certificate – IBM (May 2026)
- Agentic AI with Lang Chain and Lang Graph – IBM (May 2026)
- Develop Generative AI Applications: Get Started – IBM (May 2026)
- Oracle Cloud Infrastructure 2025 Certified DevOps Professional
- Microsoft Introduction to DevOps

ACHIEVEMENTS & RECOGNITION

- Received formal client appreciation email for zero critical post-release defects across multiple release cycles — result of designing robust pre-release validation pipelines.
- Built Python automation that reduced accessibility validation effort from 3 days to 2 hours — recognized internally for high-impact engineering that directly unblocked a compliance deadline.
- Awarded for cross-functional collaboration and automation-driven delivery improvements — worked with QA, product, and engineering to align automated testing with sprint release cadence.
- Developed and integrated RFID/NFC-based smart access and tag management solutions, implementing secure read/write, password protection, and authentication workflows for NTAG213/216 tags.
- Successfully completed the IBM RAG and Agentic AI Professional Certificate and RAG & Agentic AI Capstone Project from Coursera, gaining hands-on expertise in RAG, Agentic AI, LangChain, LangGraph, Vector Databases, and Multi-Agent Systems.